

CHAPTER 1 CHEAT SHEET — IP Static Routing & Troubleshooting (SRWE Modules 15-16)

BMIT3084 Enterprise Networking | For midterm revision | 1 page

4 Static Route Types

Standard — route to one specific network

```
ip route 192.168.1.0 255.255.255.0 10.0.0.1
```

Default — matches everything (last resort)

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

```
ipv6 route ::/0 2001:db8::1
```

Floating — backup path (higher AD)

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1 (AD 1)
```

```
ip route 0.0.0.0 0.0.0.0 10.10.10.2 5 (AD 5)
```

Summary — one route for many networks

```
ip route 192.168.0.0 255.255.252.0 s0/1/0
```

3 Next-Hop Styles

Next-hop — only IP address given

```
ip route 172.16.1.0 255.255.255.0 172.16.2.2
```

Directly connected — only exit intf

```
ip route 172.16.1.0 255.255.255.0 s0/1/0
```

Warning: serial point-to-point ONLY

Fully specified — both interface + IP

```
ip route 172.16.1.0 255.255.255.0 s0/1/0 172.16.2.2
```

Required for: Ethernet, IPv6 link-local

Host Route

One single device (/32 v4, /128 v6)

```
ip route 209.165.200.238 255.255.255.255 198.51.100.2
```

```
ipv6 route 2001:db8:acad:2::238/128 2001:db8:acad:1::2
```

3 ways: auto (local L route), static, learned

IPv6 Must-Know

ipv6 unicast-routing (enable first)

link-local next-hop? MUST be fully specified

because link-local addrs are not in routing table

Command Syntax

IPv4:

```
ip route <net> <mask> {ip | exit-intf} [distance]
```

IPv6:

```
ipv6 route <prefix/len> {ipv6 | exit-intf} [distance]
```

Administrative Distance (AD)

Lower AD = more preferred

Connected 0

Static 1 (default)

Floating 5+ (you set it)

OSPF 110

RIP 120